P.25

Program to find prime numbers between given ranges using function.

```
void main()
{
    int num, flag, count;
    int prime_check(int n);
    clrscr();
    count = 0;
    printf(" prime numbers between 10 and 50 are:\n");
    for (num = 10; num <= 50; num++)
    {
        flag = prime_check(num);
        if (flag == 1)      /* if num is prime */
        {
            printf("%5d", num);
        } count++;
    }
    printf(" Count of primes is %d ", count);
    getch();
}

int prime_check(int n)
{
    div = n/2;
    while (div > 1)
    {
        if (n%div == 0)
        {
            break;
        }
        div--;
    }
    if (div > 1)
        return(0);
    else
        return(1);
} /* function ends here */
```

P.26

program to find sum of n natural numbers using function

```
void main()
{
    int num[20], n, sum;
    int sum_nat(int arr[], int size);
    clrscr();
    printf("how many natural numbers: \n");
    scanf("%d", &n);
    printf("enter %d natural numbers: ", n);
    for(int i=0; i<n; i++)
    {
        scanf("%d", &num[i]);
    }
    sum = sum_nat(num, n);
    printf("sum of %d natural numbers is: %d", n, sum);
    getch();
}
int sum_nat(int arr[], int size)
{
    int total = 0;
    for(int i=0; i<size; i++)
    {
        total += arr[i];
    }
    return(total);
}
```

P. 28

program to convert lowcase characters to uppercase in a string.

```
void main()
{
 int i;
 printf("enter the string:");
 gets(str);
 printf("entered string is : %s \n", str);
 /* conversion to uppercase */
 for(i=0; str[i]!='\0'; i++)
 {
    if(str[i]>='a' && str[i]<='z')
        str[i] = str[i]-32;
 }
 printf("\n Modified string is : %s ", string);
 getch();
}
```

/* difference between ASCII Value of lower and uppercase alphabets is 32. Example, A ascii value is 65
a " is 97 */

P. 29

program to count the number of alphabets, digits and special characters in a string.

```
void main()
{
 char str[30];
 int alpha=0, digit=0, spchar=0, i;
 clrscr();
 printf("enter the string:");
 gets(str);
 printf("\nentered string is : %s", str);
```

```
for(i=0; str[i]!='\0'; i++)
{
  if (str[i]>='a' && str[i]<='z')
      alpha++;
  else if (str[i]>='0' && str[i]<='9')
      digit++;
  else
      spchar++;
}
printf("count of alphabets = %d \n", alpha);
printf("count of digits = %d \n", digit);
printf("count of special character = %d ", spchar);
getch();
```

/* Program ... pointers */

```
void main()
{
  char ch, *cptr;
  int x, *iptr;
  float y, *fptr;
   x= 336;
   y= 12.5;
   ch= 'J';
   cptr = &ch;
   iptr = &x;
   fptr = &y;
   clrscr();
   printf("\n value of ch= %c", *cptr);
   printf("\n address of ch is %u", cptr);
   printf("\n value of x = %d", *iptr);
   printf("\n address of x is %u", iptr);
   printf("\n value of y = %.2f", *fptr);
   printf("\n address of y is %u", fptr);
   getch();
}
```

P.30

P.31

```
/* program to perform arithmetic operations
        using pointers */

void main()
{
   int val1, val2, val3, val4, *ptr1, *ptr2;

    val1 = 100;
    val2 = 250;
    ptr1 = &val1;
    ptr2 = &val2;
    val3 = *ptr2 + *ptr1/10;          /* val3 = 250 + 100/10
                                         = 260 */
    val4 = 50 * *ptr2/*ptr1 - 20;     /* val4 = 50 * 250/100-2
                                         = 105 */
    clrscr();
   printf("\n Address of val1 is %u", ptr1);
   printf("\n Address of val2 is %u", ptr2);
   printf("\n val1 = %d  val2 = %d ", val1, val2);
   printf("\n val3 = %d  val4 = %d", val3, val4);

}
```

P. 32

Program: To swap two variables/numbers using call by reference (pointers).

```
void main()
{
    float value1, value2;
    void swap(float *a, float *b);
    clrscr();
    printf("enter two numbers:");
    scanf("%f%f", &value1, &value2);
    printf("\nBefore exchange:");
    printf("\nvalue1 = %.d, value2 = %.d", value1, value2);
    swap(&value1, &value2);
    printf("\n Exchanged values are: \n");
    printf("\n value1 = %.2f, value2 = %.2f", value1, value2);
    getch();
}

void swap(float *a, float *b)
{
    float temp;
    temp = *a;
    *a = *b;
    *b = temp;
    return;
}
```

P.33

```
/* program to demonstrate the concept of pointer
                to pointer */

void main()
{
  int val = 336, *ptr, **ptr_to_ptr;
  ptr = &val;
  ptr_to_ptr = &ptr;
  clrscr();
 printf("\n address of val is %u", &val);
 printf("\n address of val is %u", ptr);
 printf("\n address of val is %u", *ptr_to_ptr);
 printf("\n address of ptr is %u", &ptr);
 printf("\n address of ptr is %u", ptr_to_ptr);
 printf("\n address of ptr_to_ptr is %u", &ptr_to_ptr);
 printf("\n value of ptr is %u", ptr);
 printf("\n value of ptr_to_ptr is %u", ptr_to_ptr);
```

```
    printf("\n value of val is=%d", val);
    print("\n value of val is = %d", *(&val));
    printf("\n value of val = %d", *ptr);
    printf("\n value of val = %d", **ptr_to_ptr);
    getch();
}
```

P. 34

program. Program to read and print student details using structure pointers.

```
#include<stdio.h>
#include<conio.h>
typedef struct
Struct stud
{
  int roll_no;
  char name[30];
  char class[20];
  int attendance;
} student;          /* structure variable declared */

main()
{
 student * student1;
 clrscr();
 printf("enter student's record :\n");
 printf(" Roll no: ");
 scanf("%d", &student1 -> roll_no);
 printf("\n Name: ");
 gets(student1->name);
 printf("\n class: ");
 gets(student1 ->class);
 printf("\n Attendance :");
scanf("%d", &student1->attendance);
 printf("\n Entered record of student is :\n");
 printf("Roll no. %d \n Name : %s \n class : %s \n
        attendance : %d", student1 -> rollno, student1->name,
        student1->class, student1->attendance);
getch(); }
```

P. 35

program to store information and display it using copy structure.

```
#include <stdio.h>
#include <conio.h>
struct person
{
   char name[20];
   int age;
   char city[20];
} per1, per2;

void main()
{
 clrscr();
 printf("enter the record for person 1 \n\n");
 gets(per1.name);
 scanf("%d",&per1.age);
 gets(per1.city);

 per2 = per1;
 clrscr();
 printf(" entered record for person 1 : \n");
 printf(" name : %s", per1.name);
 printf(" age : %d", per1.age);
 printf(" city is : %s", per1.city);
 printf(" copied record to person2 : \n\n");
 printf(" name: %s", per2.name);
 printf(" age: %d", per2.age);
 printf(" city is %s", per2.city);
 getch();
}
```

P. 36

program to add two numbers using structure to a function structure

```
#include<stdio.h>
#include<conio.h>
struct addition
{
  int x,y;
}add;
void main()
{
  void add_num(struct addition a);  /* function declared with parameter type structure */
  clrscr();
  add_num(add);   /* function call */
  getch();
}
  /* function definition */
void add_num(struct addition a)
{
  scanf("%d",&a.x);
  scanf("%d",&a.y);
  printf("addition is %d", a.x + a.y);
}
```

P.37

program to create an array of structure for five students.

```
#include <stdio.h>
#include <conio.h>
struct student
{
   int roll_no;
   char name[30];
   char branch[30];
} stud[5];      /* array of structures defined */

main()
{
   int i
   clrscr();
   for(i=0; i<5; i++)
   {
     printf("enter the record for student - %d", i+1);
     scanf("%d",&stud[i].rollno);
     gets (stud[i].name);
     gets (stud[i].branch);
   }
     /* display the data */
    for (i=0; i<5; i++)
    {
      printf("Record of student : %d", i+1);
      printf("\n student Roll no : %d", stud[i].roll_no);
      printf("\n student name : %s", stud[i].name);
      printf("\n student branch : %s", stud[i].branch);
      getch();
    }
   getch();
}
```

P. 38

program to read text from a file.

```
void main()
{
  FILE *fptr;
  char ch;
  fptr = fopen("text.txt","r");
  clrscr();
  while((ch=getc(fptr)) != EOF)
  {
     printf("%c",ch);
  }
  fclose(fptr);
  getch();
}
```

P. 39

program to write sentence in a file.

```
void main()
{
  FILE *fptr;
  char ch;
  fptr = fopen("text.txt","w");
  printf("enter the text: \n");
  while((ch=getchar()) != EOF)
  {
     ~~putc(ch=getchar()) != EOF)~~
     putc(ch,fptr);
  }
  fclose(fptr);
}
```

P. 40

program to find the total number of lines in a file.

```
void main()
{
    FILE *fptr;
    char ch;
    int line_count=0;
    fptr = fopen("text.  ", "r");
    clrscr();
    if (!fptr)
        printf("can't open file for reading");
    else
    {   while ((ch= getc(fptr))!= EOF)
        {
            if (ch=='\n')
                line_count++;
        }
        fclose(fptr);
        printf("number of lines: %d", line_count);
    }
    getch();
}
```